

A Beginner's Guide to Solidity, the Smart Contract Programming Language

Solidity is a smart contract programming language that allows you to program decentralisation directly into every contract, ensuring transparency and security. Discover the role it plays in the Web3 ecosystem and learn how to write your first Solidity contract.



Imagine being able to create digital agreements that run themselves—no middlemen, no delays, and no need to ‘trust’ anyone. That’s exactly what smart contracts do. They’re one of the most exciting parts of blockchain technology, powering things like crypto payments, NFT trades, online voting, and even decentralised apps (DApps).

But here’s the catch: these smart contracts don’t write themselves. That’s where Solidity comes in.

Whether you’re a curious beginner, a developer interested in blockchain, or just someone who wants to understand how Web3 apps are built, this guide is for you.

By the end of this guide, you’ll have a solid understanding of:

- What Solidity is and why it’s essential for building smart contracts.
- The key role it plays in the Web3 ecosystem.
- How smart contracts function behind the scenes.
- How to write and deploy your very first Solidity contract.

What is Solidity?

Solidity is a high-level, contract-oriented programming language designed specifically for building smart contracts on the Ethereum blockchain. Created in 2014 by the Ethereum team, including co-founder Dr Gavin Wood, it was developed to bring advanced functionality to Ethereum beyond simple cryptocurrency transfers. With Solidity, developers can write self-executing contracts that run on the Ethereum Virtual Machine (EVM), enabling decentralised apps (DApps), crypto tokens, DeFi platforms, and much more.

Today, Solidity is the most widely used language in the Web3 space and is supported not just on Ethereum, but also on a growing number of EVM-compatible blockchains like Binance Smart Chain, Polygon, Avalanche, and others. It is a foundational tool for anyone looking to build in the world of decentralised technology.